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## 3. Characteristics of Remotely Sensed Data

### 3.1. Introduction

Regardless of passive or active remote sensing systems, all sensing systems detect and record energy “signals” from the Earth’s surface features or from the atmosphere. Familiar examples of remote sensing systems include aerial cameras and video recorders. Sensing systems that are more complex include electronic scanners, linear/area arrays, laser scanning systems, etc. Data collected by these remote sensing systems can be either in analog format, for example, hardcopy aerial photography or video data, or in digital format, such as a matrix of “brightness values” corresponded to the average radiance measured within an image pixel. Digital remote sensing images may be directly input into a GIS for use. Analog data can also be used in GIS through an analog-to-digital conversion or by scanning. More often, remote sensing data are first interpreted and analyzed through various methods of information extraction to provide needed data layers for GIS. The success of data collection from remotely sensed imagery requires an understanding of four basic resolution characteristics, namely, spatial, spectral, radiometric, and temporal resolution (Jensen, 2005).

### 3.2. Data Recording Methods

## 3.2.1. Photographs

As discussed in [Chap. 1](#), the output of a remote sensing system is usually an image representing the scene being observed. A remote sensing image is a 2D representation of part of the Earth's surface. Depending on the types of remote sensing devices used, remote sensing images may be analog or digital. Aerial photographs are examples of analog images, whereas satellite images typically belong to digital images.

Aerial photographs are made possible through a photographic process using chemical reactions on the surface of light-sensitive film to detect and record energy variations. Aerial photographs normally record over the wavelength range from 0.3 to 0.9  $\mu\text{m}$ , that is, from ultraviolet to visible and near-infrared (IR) spectra. The range of wavelengths to be detected by a camera is determined by the spectral sensitivities of the film. Sometimes, filters are used in the camera in conjunction with different film types to restrict the wavelengths being recorded or to reduce the effect of atmospheric scattering. Mapping cameras used to obtain aerial photographs are usually mounted in the nose or underbelly of an aircraft. Aerial photographs can also be taken from space shuttles, unmanned aviation vehicles, balloons, or even kites. A mapping camera that records data photographically requires that the film be brought back to the ground for processing and printing. [Figure 3.1](#) provides examples of black-and-white aerial photographs taken in the spring of 1998 (specifically the United States Geological Survey Digital Orthophoto Quadrangle). These samplers illustrate different land use types in Indianapolis, Indiana, U.S.A.

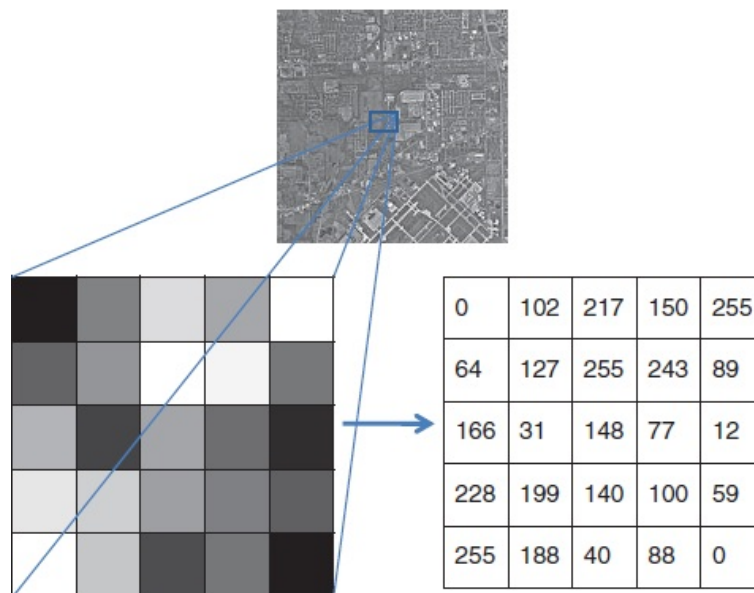
**Figure 3.1** Examples of aerial photographs showing various land-use types.



## 3.2.2. Satellite Images

Satellite imagery may be viewed as an extension of aerial photography because satellite remote sensing relies on the same physical principles to acquire, interpret, and extract information content. Satellite images are taken at a higher altitude, which allows for covering a larger area of the Earth's surface at one time, and more importantly, satellite images are acquired by electronic scanners and linear/area arrays. Satellite sensors can usually detect and record a much wider range of electromagnetic radiation, from ultraviolet radiation to microwave. The electromagnetic energy is recorded electronically as an array of numbers in digital format, resulting in digital images. In a digital image, individual picture elements, called pixels, are arranged in a 2D array in columns and rows. Each pixel has an intensity value and a location address in the 2D array, representing the electromagnetic energy received at the particular location on the Earth's surface. **Figure 3.2** illustrates pixels and how they make up digital images. The intensity value can have different ranges, depending upon the sensitivity of a sensor to the changes in the magnitude of the electromagnetic energy. In most cases, a zero value indicates no radiation is received at a particular location and the maximum recordable number of the sensor indicates the maximum detectable radiation level. For example, if a satellite sensor has set the maximum detectable radiation level to be 255, it would have a range of 256 possible intensity levels in digital number. Then, when a computer displays a digital image acquired by this sensor, 256 different brightness levels (i.e., from 0 to 255) can possibly be used.

**Figure 3.2** *Pixels—picture elements made up of digital images.*

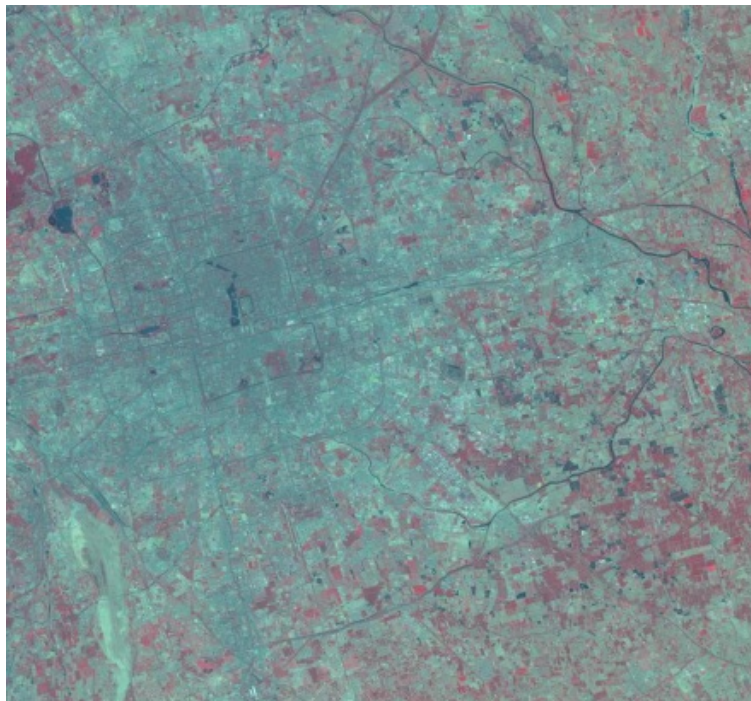


More than often, satellite sensors are designed to record the electromagnetic energy at multiple channels simultaneously. Each channel records a narrow wavelength range, sometimes known as a band. The digital image from each channel can be displayed as a black-and-white image in the computer, or we can combine and display three channels/bands of a digital image as a color image by using the three primary colors (blue, green, and red) of the computer. In that case, the digital data from each channel is represented as one of the primary colors according to the relative brightness of each pixel in that channel. Three primary colors combine in different proportions to produce a variety of colors as seen in the computer. **Figure 3.3** shows a black-and-white image of Beijing, the capital of China, and its surrounding areas. This image was taken by SPOT-5, a French satellite sensor, on June 2, 2004. **Figure 3.4** shows a false color composite of Beijing, which combines green, red, and near-IR bands of the image taken on the same date of June 2, 2004. All bands have a spatial resolution of 10 m.

**Figure 3.3** A black-and-white satellite image of Beijing, China. This image was taken by SPOT, a French satellite sensor, on June 2, 2004.



**Figure 3.4** A false color composite of Beijing, which combines green, red, and near-IR bands of the image taken on June 2, 2004.

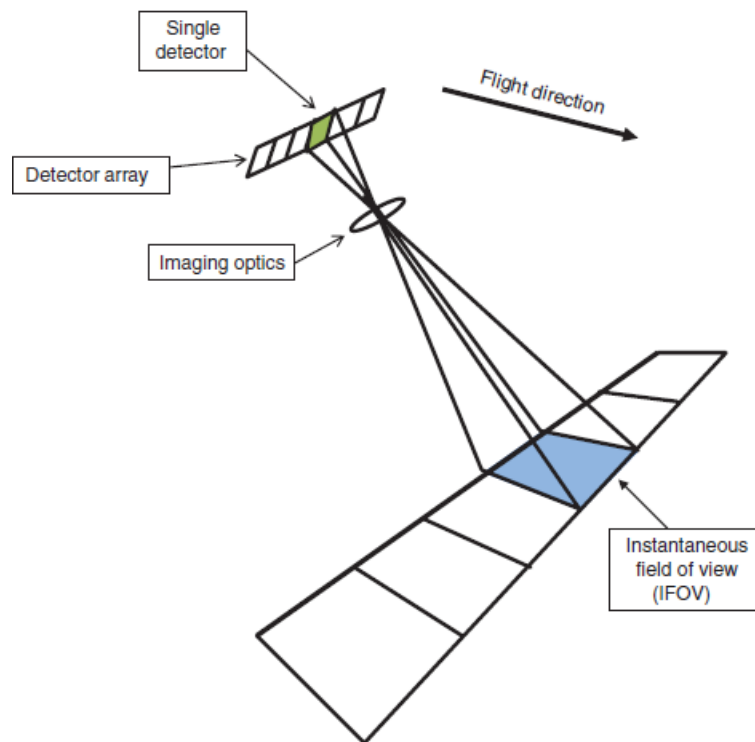


## 3.3. Spatial Resolution

Spatial resolution defines the level of spatial detail depicted in an image, and it is often defined as the size of the smallest possible feature that can be detected from an image. This definition implies that only objects larger than the spatial resolution of a sensor can be picked out from an image. However, a smaller feature may sometimes be detectable if its reflectance dominates within a particular resolution cell or it has a unique shape (e.g., linear features). Another meaning of spatial resolution is that a ground feature should be distinguishable as a separate entity in the image. However, the separation from neighbors or background is not always sufficient to identify the object. Therefore, spatial resolution involves both ideas of detectability and separability. For any feature to be resolvable in an image, it involves consideration of spatial resolution and spectral contrast, as well as feature shape.

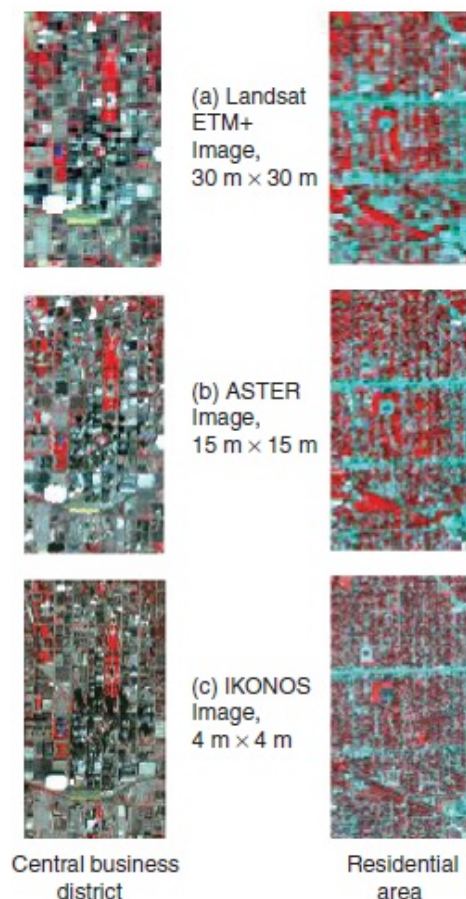
Spatial resolution is a function of sensor altitude, detector size, focal size, and system configuration (Jensen, 2005). For aerial photography, spatial resolution is measured in resolvable line pairs per millimeter; while for other sensors, it refers to the dimensions (in meters) of the ground area that falls within the instantaneous field of view (IFOV) of a single detector within an array (Jensen, 2005). **Figure 3.5** illustrates the concept of IFOV of a sensor as a measure of the ground area viewed by a single detector element in a given instant in time. The technical details of spatial resolution for aerial photographs are explored further in **Chap. 4**.

**Figure 3.5** The concept of instantaneous field of view (IFOV).



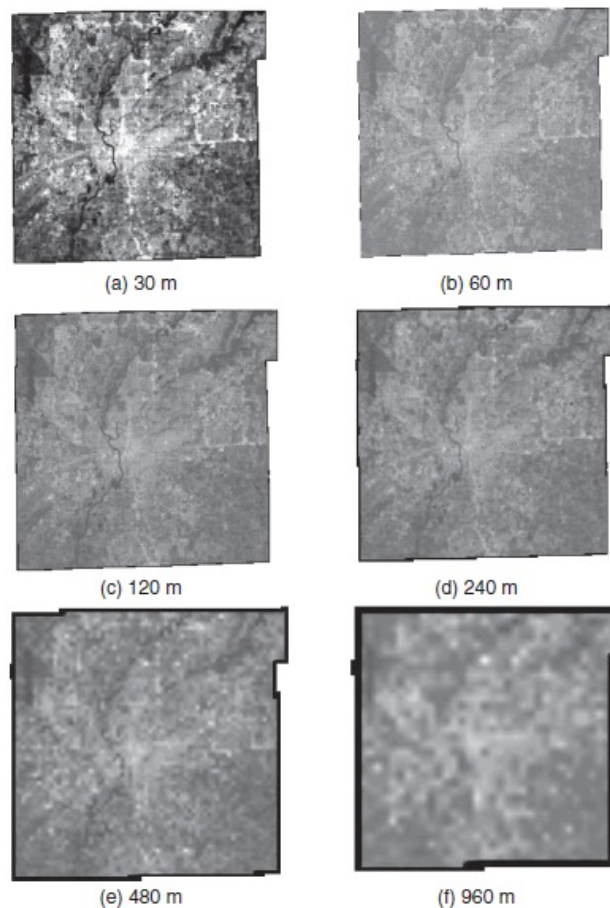
Spatial resolution determines the level of spatial details that can be observed on the Earth's surface. Coarse spatial-resolution images can detect only large features, while with fine resolution images, small features become visible. Figure 3.6 gives a comparison of a Landsat ETM+ image (spatial resolution = 30 m), an ASTER image (spatial resolution = 15 m), and an IKONOS image (spatial resolution = 4 m). These images were taken between 2000 and 2003 over the downtown area and a residential area in Indianapolis, Indiana. The IKONOS image gives fine details for each building in the downtown area and roads, houses, and vegetation in the residential area that Landsat and ASTER images cannot. However, for a remote sensing project, image spatial resolution is not the only factor needed to be considered. The relationship between the geographical scale of a study area and the spatial resolution of remote sensing image has to be studied (Quattrochi and Goodchild, 1997). At the local scale, high spatial-resolution imagery, such as IKONOS and QuickBird data, is more effective. At the regional scale, medium spatial-resolution data, such as Landsat TM/ETM+ and Terra ASTER data, are the most frequently used. At the continental or global scale, coarse spatial-resolution data, such as AVHRR and MODIS data, are most suitable. Higher resolution means the need for larger data storage and higher cost, and may introduce difficulties in image processing for a large study area. One goal for satellite sensors since the late 1990s has been improved spatial resolution, which now is down to better than 1 m. Sub-meter satellite images have introduced a new issue of residential privacy and have resulted in some legal cases.

**Figure 3.6** Comparison of spatial resolution among Landsat ETM+, ASTER, and IKONOS images.



As discussed previously, digital images are composed of pixels. It is important not to confuse the spatial resolution of a digital image and pixel size, although the two concepts are interrelated. In Fig. 3.7(a), a Landsat ETM+ image covering the city of Indianapolis, Indiana, United States, is displayed at full resolution. The spatial resolution of Landsat reflected bands is 30 m, and in Fig. 3.7(a), each pixel represents an area of 30 m × 30 m on the ground. In this case, the spatial resolution is the same as the pixel size of the image. In Fig. 3.7(b) to (f), the Landsat image is re-sampled to larger pixel sizes, ranging from 60 to 960 m. At the pixel size of 30 m, roads, rivers, and a central business district are visible. As the pixel size becomes larger, these features and objects become indiscernible. As discussed previously, pixel size is primarily defined by IFOV. This “optical pixel” is generally a circular or elliptical shape and combines radiation from all components within its ground area to produce the single pixel value in a given spectral channel (Harrison and Jupp, 2000). Along each image line, optical pixel values are sampled and recorded, and the average spacing of optical pixels relative to the ground area covered is then used to determine the geometric pixel size by using various mathematical algorithms (Harrison and Jupp, 2000). Coarse resolution images may include a large number of mixed pixels, where more than one surface cover type (several different features/objects) can be found within a pixel. Fine spatial resolution data considerably reduce the mixed pixel problem.

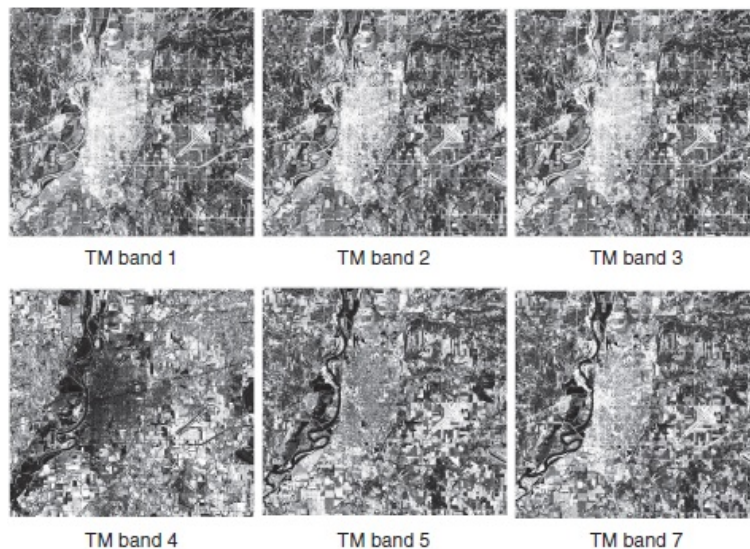
**Figure 3.7** Effect of digitizing the same area with different pixel sizes.



## 3.4. Spectral Resolution

In [Chap. 2](#), we discussed the concept of spectral signature. Different features/materials on the Earth's surface interact with electromagnetic energy in different and often distinctive ways. A specific spectral response curve, or spectral signature, can be determined for each material type. Broad classes, such as vegetation, water, and soil, in an image can often be distinguished by comparing their responses over distinct, very broad wavelength ranges. Spectral signature, or spectral response curve, is related to the spectral resolution of a remote sensor. Each remote sensor is unique with regard to what portion of the electromagnetic spectrum it detects and records. Moreover, each remote sensor records different numbers of segments of the electromagnetic spectrum, or bands, which may also have different bandwidths. Spectral resolution of a sensor, by definition, refers to the number and the size of the bands it is able to record (Jensen, 2005). For example, AVHRR, Onboard NOAA's Polar Orbiting Environmental Satellite (POES) platform, collects four or five broad spectral bands (depending upon individual instrument) in visible (0.58 to 0.68  $\mu\text{m}$ , red), near-IR (0.725 to 1.1  $\mu\text{m}$ ), mid-IR (3.55 to 3.93  $\mu\text{m}$ ), and thermal IR portions (10.3 to 11.3  $\mu\text{m}$  and 11.5 to 12.5  $\mu\text{m}$ ) of the electromagnetic spectrum. AVHRR, acquiring image data at the spatial resolution of 1.1 km at nadir, has been extensively used for meteorological studies, vegetation pattern analysis, and global modeling. Landsat Thematic Mapper (TM) sensor collects seven spectral bands. Its spectral resolution is higher than early instruments onboard Landsat such as MSS and RB). [Figure 3.8](#) illustrates all reflective (excluding thermal IR band) bands of a Landsat TM image of Terre Haute, Indiana, U.S.A.

**Figure 3.8** An example of Landsat TM spectral bands.



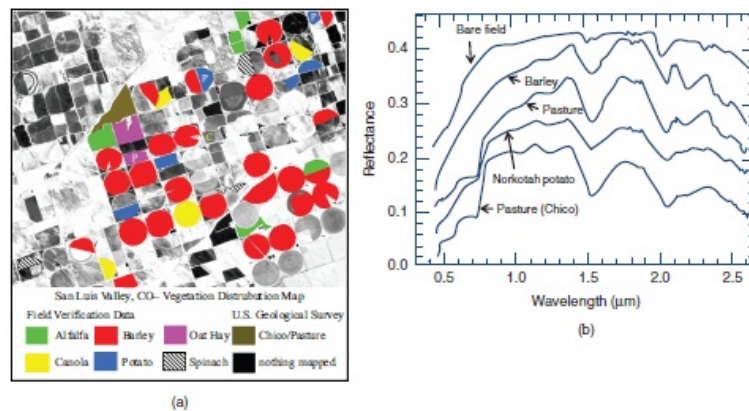
Most satellite sensors are engineered to use particular electromagnetic channels to differentiate between major cover types of interest, or materials, on the Earth's surface. The position, width, and number of spectral channels being sensed have been determined before a sensor is carried into the space. [Table 3.1](#) shows the usages of each of the seven spectral bands of Landsat TM sensor. Spectral extent describes the range of wavelengths being sensed in all channels of an image. A remote sensing system may detect electromagnetic radiation in the visible and near IR regions, or extend to the middle and thermal IR regions. Microwave radiation may be recorded using radar and passive microwave sensors. An increase in spectral resolution over a given spectral extent implies that a greater number of spectral channels are recorded. The finer the spectral resolution, the narrower the range of wavelength for a particular channel or band. The increased spectral resolution allows for a fine discrimination between different targets, but also increases costs in the sensing system, and data processing on board and on the ground. Therefore, the optimal spectral range and resolution for a particular cover type need to be modified with respect to practical uses of data collection and processing.

**Table 3.1** Landsat Thematic Mapper Bands and Usages

| Name | Wavelength Range (nm) | Application                                                                                                                        |
|------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| TM 1 | 0.45–0.52 (blue)      | soil/vegetation discrimination; bathymetry/ coastal mapping; cultural/urban feature identification                                 |
| TM 2 | 0.52–0.60 (green)     | green vegetation mapping (measures reflectance peak); cultural/urban feature identification                                        |
| TM 3 | 0.63–0.69 (red)       | vegetated vs. non-vegetated and plant species discrimination (plant chlorophyll absorption); cultural/urban feature identification |
| TM 4 | 0.76–0.90 (near IR)   | identification of plant/vegetation types, health, and biomass content; water body delineation; soil moisture                       |
| TM 5 | 1.55–1.75 (short IR)  | sensitive to moisture in soil and vegetation; discriminating snow and cloud-covered areas                                          |
| TM 6 | 10.4–12.5 (TIR)       | vegetation stress and soil moisture discrimination related to thermal radiation; thermal mapping (urban, water)                    |
| TM 7 | 2.08–2.35 (short IR)  | discrimination of mineral and rock types; sensitive to vegetation moisture content                                                 |

A major advance in remote sensor technology in recent years has been a significant improvement in spectral resolution, from bandwidths of tens of hundreds of nanometers (1 nm = 1000 nm), as pertains to the Landsat TM, to 10 nm or less. Hyperspectral sensors (imaging spectrometers) are the kind of instruments that acquire images in many very narrow, contiguous spectral bands throughout the visible, near-IR, mid-IR, and thermal-IR portions of the spectrum. Whereas Landsat TM obtains only one data point corresponding to the integrated response over a spectral band 0.24-nm wide, a hyperspectral sensor, for example, is capable of obtaining 24 data points over this range using bands approximately 0.01-nm wide. NASA Airborne Visible-Infrared Imaging Spectrometer (AVIRIS) collects 224 contiguous bands with wavelengths from 400 to 2500 nm. A broadband system can only discriminate general differences among material types, while a hyperspectral sensor affords the potential for detailed identification of materials and better estimates of their abundance. [Figure 3.9\(a\)](#) shows an AVIRIS image of some circular fields in the San Juan Valley of Colorado. Vegetation or crop types in the fields, identified by different colors in the figure, are determined based on field survey and the spectral curves plotted in [Fig. 3.9\(b\)](#) for the crops indicated. Another example of a hyperspectral sensor is MODIS, on both NASA's Terra and Aqua missions and its follow-on to provide comprehensive data about land, ocean, and atmospheric processes simultaneously. MODIS has a two-day repeat global coverage with spatial resolution (250, 500, or 1000 m depending on wavelength) in 36 spectral bands.

**Figure 3.9** Vegetation species identification using hyperspectral imaging technique. (a) AVIRIS image showing circular fields in San Luis Valley, Colorado, and (b) reference spectra used in the mapping of vegetation species (From Clark et al. 1995, with permission).

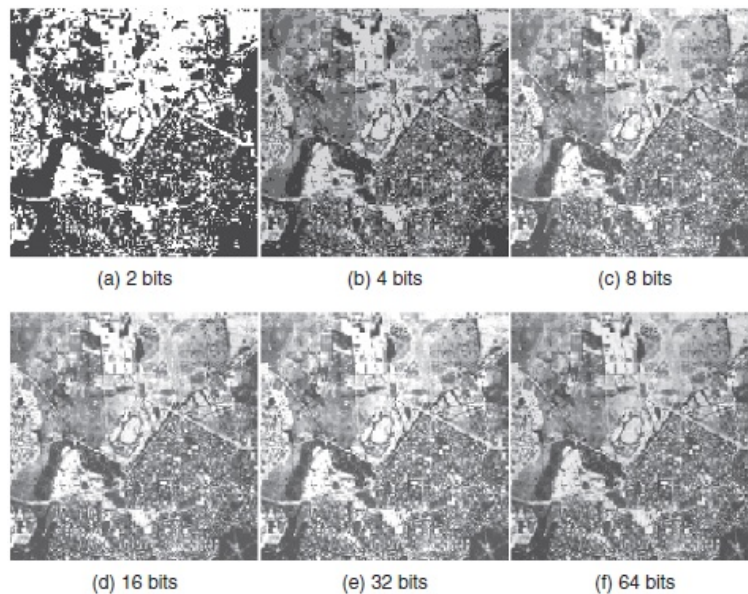


## 3.5. Radiometric Resolution

Radiometric resolution refers to the sensitivity of a sensor to incoming radiance, that is, how much change in radiance there must be on the sensor before a change in recorded brightness value takes place (Jensen, 2005). Therefore, radiometric resolution describes a sensor's ability to discriminate subtle differences in the detected energy. Coarse radiometric resolution would record a scene using only a few brightness levels, that is, at very high contrast, whereas fine radiometric resolution would record the same scene using many brightness levels. Radiometric resolution in digital imagery is comparable to the number of tones in an aerial photograph (Harrison and Jupp, 2000). For both types of remote sensing images, radiometric resolution is related to the contrast in an image.

Radiometric resolution is frequently represented by the levels of quantization that is used to digitize the continuous intensity value of the electromagnetic energy recorded. Digital numbers in an image have a range from 0 to a selected power of 2 minus 1. This range corresponds to the number of bits (binary digits) used for coding numbers in computers. For example, Landsat-1 MSS initially records radiant energy in six bits (values ranging from 0 to 63, i.e., levels of quantization =  $64 = 2^6$ ), and was later expanded to seven bits (values ranging from 0 to 127, equal to  $2^7$ ). In contrast, Landsat TM data are recorded in eight bits, that is, its brightness levels range from 0 to 255 (equal to  $2^8$ ). The effect of changes in radiometric resolution on image feature contrast is illustrated in Fig. 3.10. Because human eyes can only perceive 20 to 30 different grey levels, the additional resolution levels provided by digital images are not visually discernible (Harrison and Jupp, 2000). Digital image processing techniques using computers are necessary in order to derive maximum discrimination from the available radiometric resolution.

**Figure 3.10** Effect of changes in radiometric resolution on image feature contrast (From Harrison and Jupp, 2000).



## 3.6. Temporal Resolution

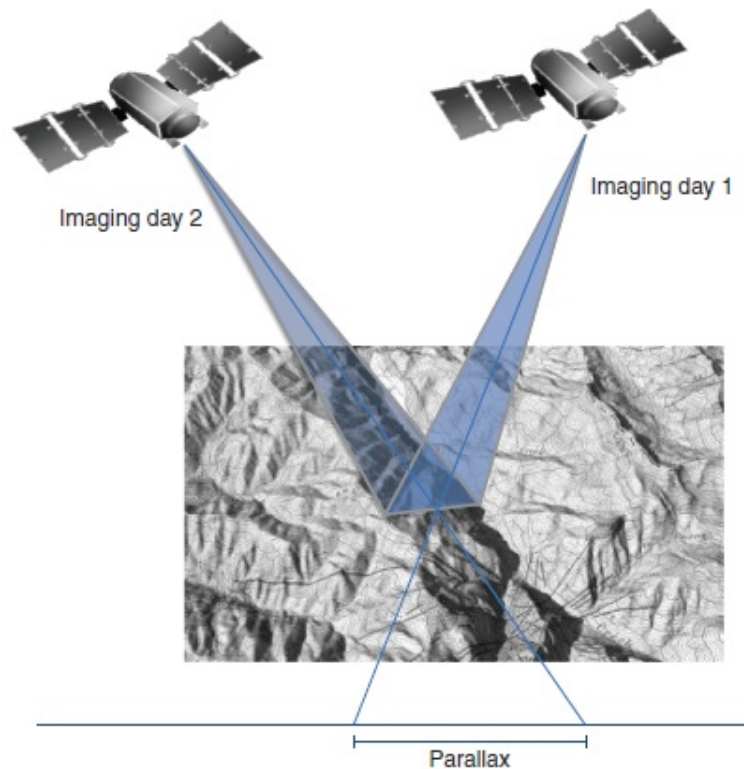
Temporal resolution refers to the amount of time it takes for a sensor to return to a previously imaged location, commonly known as the repeat cycle or the time interval between acquisitions of two successive images. For satellite remote sensing, the length of time for a satellite to complete one entire orbital cycle defines its temporal resolution. Depending on the types of satellites, the revisit time may range from 15 min to more than 10 days. For air-borne remote sensing, temporal resolution is less pertinent because users can schedule flights for themselves.

Temporal resolution has an important implication in change detection and environmental monitoring. Many environmental phenomena constantly change over time, such as vegetation, weather, forest fire, volcano, and so on. These changes are reflected in the spectral characteristics of the Earth's surface, and are recorded in remote sensing images. Temporal differences between remotely sensed imagery are caused not only by the changes in spectral properties of the Earth's surface features/objects, but also they can result from atmospheric differences and changes in the sun's position during the course of a day and during the year. Temporal resolution is an important consideration in remote sensing of vegetation because vegetation grows according to daily, seasonal, and annual phenological cycles. It is crucial to obtain anniversary or near-anniversary images in change detection of vegetation. Anniversary images greatly minimize the effect of seasonal differences (Jensen, 2005). Many weather sensors have a high temporal resolution: GOES (every 15 min), NOAA AVHRR local area coverage (2 times per day), and Meteosat first generation (every 30 min). Observing short-lived, time-sensitive phenomena, such as floods, fire, and oil spills, requires truly high temporal resolution imagery. In general, ideal temporal resolution varies considerably for different applications. While monitoring of crop vigor and health may require daily images in a growing season, studies of urban development only need to have an image every 1 to 5 years (Jensen, 2007). Monitoring Earth's ecosystems, especially forests, is the major goal of many satellite missions. Human activities and natural forces can cause forest disturbance. Harvest, fire, and storm damage often result in abrupt changes, while changes due to insects and disease can last several years or longer. However, tree growth is always a slow, gradual process. Characterization of forest growth and disturbance therefore requires remote sensing data of different temporal resolutions (Huang et al., 2010).

The set overpass times of satellites may coincide with clouds or poor weather conditions. This is especially true in tropical areas, where persistent clouds and rains in the wet season offer limited clear views of the Earth's surface and thus prevent us from getting good quality images. Moreover, some projects require remote sensing images of particular seasons. In agriculture, phenology manifests itself through the local crop calendar—the seasonal cycle of plowing, planting, emergence, growth, maturity, harvest, and fallow (Campbell, 2007). The crop calendar stipulates us to acquire satellite images in the growth season, often from late spring to fall in the mid-latitude region, but the local climate needs to be considered. On the contrary, detection of urban buildings and roads may well be suited in the leave-off season in the temperate regions. In sum, the date of acquisition should be determined to ensure the extraction of maximum information content from remotely sensed data. When fieldwork needs to coincide with image acquisition, the date of acquisition must be planned well ahead of the time.

The off-nadir imaging capability of the SPOT-5 sensor reduces the usual revisit time of 26 days to 2 to 3 days depending on the latitude of the imaged areas. This feature is designed for taking stereoscopic images and for producing digital elevation models (Fig. 3.11). However, it obviously also allows for daily coverage of selected regions for short periods, and provides another means for monitoring dynamic events such as flood or fire. The ASTER sensor has a similar capability of off-nadir imaging for its near infrared band (0.76 to 0.86  $\mu\text{m}$ ) by equipping a backward-looking telescope to acquire a stereo pair image.

**Figure 3.11** Off-nadir imaging for stereoscopic acquisitions.



## 3.7. Summary

To understand the characteristics of remotely sensed data, we must learn about the types of sensors used to retrieve these data and their functions and performance. In this chapter, we categorize remote sensors into two broad classes: analog and digital sensors. Aerial photographs, as an example of analog image, have the clear advantage of recording extremely fine spatial details. Satellite imagery, as an example of digital imagery, tends to have better quality in spectral, radiometric, and temporal resolution. The trend over the past 40 years has been directed toward improved resolution of each type. Considerable efforts have been made to design and construct sensors that maximize the resolution needed for their intended tasks. Superior spatial resolution permits ever-smaller targets, less than 1 m in size, to be seen as individuals. Greater spectral resolution means that individual entities (features, objects) can be identified more accurately because their spectral signatures are more distinguished as in hyperspectral sensors. Increased radiometric resolution offers sharper images, allowing for better separation of different target materials from their backgrounds. Finally, improved temporal resolution has made temporal analysis possible not only in the area of monitoring ecosystems and natural hazards, but also in detection of movement of ships and vehicles.

Under many situations, clear trade-offs exist between different forms of resolutions. For example, in traditional photographic emulsions, increases in spatial resolution are based on decreased size of film grain, which produces accompanying decreases in radiometric resolution, that is, the decreased sizes of grains in the emulsion portray a lower range of brightness values (Campbell, 2007). In multi-spectral scanning systems, an increase in spatial resolution requires a smaller IFOV, and thus less energy reaches the sensor. This effect may be compensated for by broadening the spectral window to pass more energy, that is, decrease spectral resolution, or by dividing the energy into fewer brightness levels, that is, decreasing radiometric resolution (Campbell, 2007). To overcome the conflict between spatial and temporal resolutions, NASA and Europe Space Agency have reportedly proposed to develop remote sensing systems that are able to provide data combining both high spatial resolution and revisit capabilities, among others.

## 3.8. Key Concepts and Terms

**Aerial photographs** Remote sensing images in analog form made possible through the photographic process using chemical reactions on the surface of light-sensitive film to detect and record energy variations. Aerial photographs normally record over the wavelength range from 0.3 to 0.9 mm, that is, from ultraviolet to visible and near-IR spectra.

**Image** Any pictorial representation, regardless of what wavelengths or remote sensing device has been used to detect and record the electromagnetic energy.

**Image bands or channels** Information from a narrow wavelength range is gathered and stored in a channel, also sometimes referred to as a band.

**Instantaneous field of view (IFOV)** A measure of the ground area viewed by a single detector element in a given instant in time.

**Pixel** Picture element that has a single brightness value or digital number in an image, representing the electromagnetic energy received at the particular location on the Earth's surface. Pixel has both spatial and spectral property.

**Radiometric resolution** The sensitivity of a sensor to incoming radiance, and describes a sensor's ability to discriminate subtle differences in the detected energy.

**Spatial resolution** The size of the smallest object that can be resolved on the ground. In a digital image, the resolution is limited by the pixel size, that is, the smallest resolvable object cannot be smaller than the pixel size.

**Spectral resolution** Each remote sensor is unique with regard to what portion of the electromagnetic spectrum it detects and records and the number of bands and their widths.

**Temporal resolution** The amount of time it takes for a sensor to return to a previously imaged location, commonly known as the repeat cycle or the time interval between acquisitions of two successive images.

## 3.9. Review Questions

1. What are analog and digital images? How do aerial photographs differ from satellite images in respect to data recording method and the use of electromagnetic radiation energy?
2. Explain the relationship between pixels and images bands.
3. What is spatial, spectral, radiometric, and temporal resolution? Why do trade-offs exist between different forms of resolutions?
4. Why is it important to develop new sensors that improve more than one type of resolution?
5. Aerial photographs taken from airplanes usually have a clearer view of ground targets than digital images from satellite observations. For applications in agriculture and in military, respectively, explain the advantages and disadvantages of each type of remotely sensed data.
6. TM is a sensor of the Landsat program. How many spectral bands does it acquire and what are they? What is spatial resolution for its six reflected bands and that for the emitted band?
7. The pixel size of Landsat TM imagery is  $30\text{ m} \times 30\text{ m}$ , and the dimension of a complete scene is  $185\text{ km} \times 185\text{ km}$ . Calculate the number of pixels for a single band of a TM scene. Furthermore, calculate the number of pixels required to represent a single band of a Landsat MSS (79 m resolution) and make a comparison with Landsat TM imagery.
8. Which images have a better visual appearance, Landsat TM or IKONOS images? If you have a Landsat TM and an IKONOS image covering exactly the same area, which image would have a bigger file size when stored in computers?
9. Hyperspectral sensors can detect and record electromagnetic radiation in hundreds of very narrow spectral bands. What would be some of the advantages of these types of sensors? Any disadvantages?
10. A digital image has a radiometric resolution of 4 bits. What is the maximum value of the digital number that can be represented in that image? How about 6 and 12 bits?
11. Can you provide a couple of examples of using satellite sensors that have both high temporal-resolution (e.g., every 5 min) and high spatial-resolution (e.g., 1 m)?

## 3.10. References

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